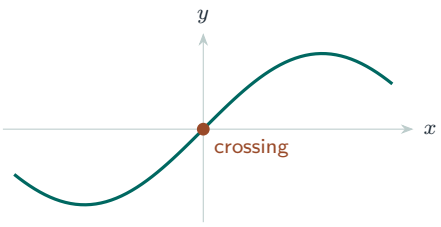
What Is So Special About Special Functions?

A wave, two coordinate charts, and the Riccati equation

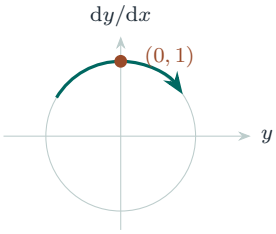
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Abstract The logarithmic derivative puts the classical scalar second-order linear equations into one projective flow. We derive interval recovery, follow a Bessel mode between two scales, and calculate the Schwarzian cost of changing coordinates. A coupled cone separates the roots of a frozen quadratic from the true exponents of the solution. The accompanying Lean proof and calculator make these distinctions reproducible.

Watch the wave $y(x) = \sin x$ cross zero. Its height vanishes, its velocity is one, and a pencil tracing the curve sails straight through. Now ask how fast the wave is changing *relative to its height*. The answer is $\cot x$. Approach the crossing from opposite sides and the answer runs off the page in opposite directions. The wave and the pencil agree that nothing dramatic happened. What did the division see?



The wave $y = \sin x$



The state $(\sin x, \cos x)$

The quotient saw a vertical direction. Writing $v = \mathrm{d}y/\mathrm{d}x$, the state vector is (y, v). Its direction can be recorded by $z = v/y$ or, near this crossing, by $w = y/v = \tan x$. The second coordinate takes the perfectly ordinary value zero. The same switch works for a complex-valued solution, with the two quotients serving as charts on the complex projective line.

The equation behind the examples

On an interval of ordinary points, take a scalar homogeneous equation

$$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} + P(x)\frac{\mathrm{d}y}{\mathrm{d}x} + Q(x)y = 0. \tag{1}$$

The quotient rule gives an identity even before we impose the equation. Primes in compact formulas denote differentiation in the displayed independent variable.

$$\underbrace{z' + z^2 + Pz + Q}_{\text{Riccati residual}} = \frac{y'' + Py' + Qy}{y}, \qquad z = \frac{y'}{y}, \quad y \neq 0. \tag{2}$$

This is the common calculation for Euler, Bessel, Airy, Hermite and Legendre. What survives the division? When does a change of coordinate give a basis of solutions? And what should a computer hand back when all it has done is rewrite the equation?

01 RECOVERY

What the wave forgot

Multiplying y by any nonzero constant changes its scale and leaves y'/y untouched. One measurement restores precisely that missing information. If $z = y'/y$ is continuous on a zero-free interval containing a, then

$$y(x) = y(a)\exp\left(\int_a^x z(t)\,\mathrm{d}t\right). \tag{3}$$

Differentiate the right side once to obtain $y' = zy$ and again to obtain $y'' = (z' + z^2)y$. Equation (1) follows whenever z satisfies its Riccati equation. Conversely, the quotient of a given nonvanishing solution and the exponential in (3) has zero derivative and equals $y(a)$ at a. No choice of complex logarithm enters this argument.

At a simple zero, differentiating the reciprocal chart gives

$$\frac{\mathrm{d}w}{\mathrm{d}x} = 1 + P(x)w + Q(x)w^2, \qquad w = \frac{y}{y'}.$$

In particular $w' = 1$ at the crossing. With continuous P, Q, uniqueness excludes $y = y' = 0$ for a nontrivial solution at an ordinary point. Thus the two charts cover every such state. The identically zero solution is known separately.

When a direction stays put

For $y'' - 3y' + 2y = 0$, the slope obeys $z' = -(z-1)(z-2)$. At either root the direction of (y, y') stays put while its length changes. Recovery gives e^x and e^{2x}. Their Wronskian is e^{3x}, so arbitrary initial height and velocity can be fitted by a unique linear combination. A mixture usually has a *moving* slope.

What if the roots coincide? For $y'' - 2y' + y = 0$, the slope equation is $z' = -(z-1)^2$. It offers one constant slope and the nonconstant solution $z = 1 + 1/x$ on $x > 0$. Recovery gives e^x and xe^x. Extending both solutions through $x = 0$ is legitimate, and their Wronskian there is one. The second mode is what a solver misses if it simply counts distinct quadratic roots.

Euler and Bessel share a clock

Write the shifted Euler variable as $t = 1 + u > 0$, or take $t = x > 0$ for Bessel. Both fit $t^2 y_{tt} + t y_t + (kt^2 - c)y = 0$. Let $s = \log t$ and $W(s) = t\,y_t/y$. Since $\mathrm{d}/\mathrm{d}s = t\,\mathrm{d}/\mathrm{d}t$, the same product rule gives both rows.

Equation	Parameters	Exact slope law in s
Shifted Euler	$k = 0,\ c = 2\ell^2$	$\mathrm{d}W/\mathrm{d}s = 2\ell^2 - W^2$
Bessel, unit argument	$k = 1,\ c = \nu^2$	$\mathrm{d}W/\mathrm{d}s = \nu^2 - W^2 - e^{2s}$

Euler has constant slopes $W = \pm\sqrt{2}\ell$. Bessel has one additional term. As $s \to -\infty$, that term tends to zero and the limiting equilibria obey $W^2 = \nu^2$. This explains the indicial exponents $\pm\nu$. It leaves room for a logarithmic second solution when the indicial roots coincide or resonate.

02 THE CLASSICAL FAMILIES

Which wave are we following?

A differential equation specifies a two-dimensional solution space. A familiar name such as J_ν or Ai picks a member of that space. A boundary condition, a normalization or an asymptotic choice does that extra work. The coefficient pair below can be read straight from the equation and inserted into (2).

Equation	$P(x)$	$Q(x)$	Detail that matters
Constant coefficients	p	q	A double root supplies xe^{rx}.
Harmonic oscillator	0	ω^2	$e^{\pm i\omega x}$ have fixed slopes.
Shifted Euler	$1/(1+x)$	$-2\ell^2/(1+x)^2$	Use a chart on either side of -1.
Bessel	$1/x$	$1-\nu^2/x^2$	Positive real zeros distinguish branches.
Airy	0	$-x$	Frozen roots meet at the turning point.
Hermite	$-2x$	$2n$	Here $y=H_n$ for the polynomial solution.
Legendre	$-2x/(1-x^2)$	$\ell(\ell+1)/(1-x^2)$	The endpoints ±1 are singular.

A standing wave and an outgoing wave

For real order $\nu \geq 0$, the regular Bessel solution behaves like a multiple of x^ν as $x \downarrow 0$. Far along the positive real axis it oscillates and has infinitely many simple zeros. Its real logarithmic derivative J_ν'/J_ν has poles there. The outgoing Hankel solution $H_\nu^{(1)} = J_\nu + iY_\nu$ makes a different choice. The Wronskian $J_\nu Y_\nu' - J_\nu' Y_\nu = 2/(\pi x)$ prevents J_ν, Y_ν from vanishing together. Hence the outgoing slope is regular for $x > 0$, and the large-x expansion gives

$$\frac{(H_\nu^{(1)})'}{H_\nu^{(1)}} = i - \frac{1}{2x} + O(x^{-2}).$$

The imaginary part records the phase advance. The small real part records the decay of the $x^{-1/2}$ envelope. Both features fit in one complex number. The asymptotics and zero statements here are classical [2].

A quick test for a moving root

Airy's equation gives $z' = x - z^2$. At $x > 0$, freeze x and the roots are $\pm\sqrt{x}$. Try $z = \sqrt{x}$ in the actual equation. The right side is zero, while the derivative on the left is $1/(2\sqrt{x})$. Freezing can suggest a leading balance. Substitution tells you whether the candidate solves the equation. The decaying real Ai branch has slope asymptotic to $-\sqrt{x}$ as $x \to +\infty$. On the negative real axis its zeros interrupt this affine chart [3].

Hermite supplies a quieter version of the same warning. The Gaussian-weighted function $e^{-x^2/2}H_n(x)$ and the polynomial $H_n(x)$ have logarithmic derivatives differing by $-x$. A claim about the large-x slope needs to say which function the symbol y denotes.

For Legendre, take $P_1(x) = x$. The coefficients are singular at $x = \pm1$, where this particular solution is regular. Its logarithmic derivative $1/x$ has a pole at the ordinary point $x = 0$. A pole in the slope and a singular point of the input equation can live at entirely different addresses.

03 A CHANGE OF COORDINATE

Where the quarter comes from

Start with the shifted Euler equation from page 2. It contains the number $2\ell^2$. After one change of dependent variable, the number $1/4$ appears alongside it. After changing the independent variable as well, the quarter disappears. Following it through both steps is a useful way to remember the signs.

First remove the drift

For differentiable P, set $y = \mu h$ with $\mu'/\mu = -P/2$. The coefficient of h' in $y'' + Py' + Qy$ is $2\mu' + P\mu = 0$. The equation that remains is

$$h'' + q(x)h = 0, \qquad q = Q - \tfrac{1}{2}P' - \tfrac{1}{4}P^2.$$

Equivalently, the slope of h is $z + P/2$. In Hermite's case, $\mu = e^{x^2/2}$, which accounts for the Gaussian on the previous page.

Then change the clock

Let $s = s(x)$ be C^3 with $s' > 0$ on the interval. Use $h(x) = (s')^{-1/2}H(s(x))$. Twice differentiating the product gives $H_{ss} + \widetilde{q}(s)H = 0$, where

$$\widetilde{q}(s(x)) = \frac{q(x) - \frac{1}{2}\{s, x\}}{(s')^2}, \qquad \{s, x\} = \frac{s'''}{s'} - \frac{3}{2}\left(\frac{s''}{s'}\right)^2. \tag{4}$$

The prefactor kills the H_s term. Indeed for $\rho = (s')^{-1/2}$ its coefficient is $2\rho' s' + \rho s'' = 0$, and $\rho''/\rho = -\{s, x\}/2$. A proposed coordinate gives a constant transformed potential c precisely when $q = c(s')^2 + \{s, x\}/2$ [1].

For $t = 1 + u > 0$, Euler has $P = 1/t$ and $Q = -2\ell^2/t^2$. Its normal-form potential and the logarithmic coordinate cost are

$$q = \frac{1/4 - 2\ell^2}{t^2}, \qquad \tfrac{1}{2}\{\log t, u\} = \tfrac{1}{2}\left(\frac{2}{t^2} - \frac{3}{2t^2}\right) = \frac{1}{4t^2}.$$

Subtract the second expression from the first and divide by $(\mathrm{d}s/\mathrm{d}u)^2 = 1/t^2$. The answer is $-2\ell^2$, exactly as on page 2. This is the logarithmic-coordinate form of the Langer quarter-term [4].

What counts as finding a coordinate?

If two independent real normal-form solutions h_1, h_2 are already known, their ratio $s = h_1/h_2$ satisfies $\{s, x\} = 2q$ on a zero-free patch of h_2. In fact s' is a nonzero constant times h_2^{-2}, and differentiating its logarithmic derivative gives the identity. Choose the sign for $s' > 0$. Equation (4) then gives $\widetilde{q} = 0$. For a *solver*, the interesting question is which coordinates can be constructed from the input coefficients before the solutions are known. The examples motivate that question. A global classification requires a specified class of admissible coordinates and a domain.

04 A COUPLED EXAMPLE

The root that nearly got away

The difference between a frozen root and a solution can be only one term. On a cone it is small enough to miss and large enough to change every nonzero exponent.

Write $t = 1 + u > 0$. Consider the metric

$$\mathrm{d}s^2 = 2\,\mathrm{d}u^2 + t^2(\mathrm{d}v + 2\pi m\,\mathrm{d}u)^2, \qquad m \in \mathbb{Z}_{>0}.$$

The coordinate $v' = v + 2\pi m u$ makes it diagonal. For a harmonic function $F(u)e^{i\ell v'}$, $\ell \in \mathbb{Z}$, the radial Laplace equation in this frame is $F'' + F'/t - 2\ell^2 F/t^2 = 0$. Returning to v multiplies F by $e^{\kappa u}$, with $\kappa = 2\pi i \ell m$. Thus $f = e^{\kappa u}F$ satisfies

$$f'' + \left(\frac{1}{t} - 2\kappa\right) f' + \left(\kappa^2 - \frac{\kappa}{t} - \frac{2\ell^2}{t^2}\right) f = 0.$$

The phase is present throughout the interval. To see exactly what it contributes, insert $z = \kappa + r/t$ into the two residuals.

Calculation	Result
Freeze the coefficients and evaluate $z^2 + Pz + Q$	$(r^2 + r - 2\ell^2)/t^2$
Differentiate the proposed slope	$z' = -r/t^2$
Evaluate the full Riccati residual $z' + z^2 + Pz + Q$	$(r^2 - 2\ell^2)/t^2$

On resonance $\ell = m$, the two frozen roots have real offsets $r = (-1 \pm \sqrt{1 + 8m^2})/2$ and the same imaginary part $2\pi m^2$. Integrating those offsets as though they were actual slopes would leave a nonzero residual. The derivative of r/t supplies the missing term.

Exact modes for $\ell \neq 0$ $\qquad f_\pm(u) = (1+u)^{\pm\sqrt{2}\ell}\, e^{2\pi i \ell m u}.$

The two exponents are distinct, so both radial modes survive multiplication by the common nonzero phase. For $\ell = 0$, the repeated exponent gives 1 and $\log(1 + u)$. This agrees with separation in the diagonal frame. Try $\ell = m = 1$. The positive frozen offset is $r = 1$, which would give $f = (1 + u)e^{2\pi i u}$. At $u = 0$, substitution into the linear equation returns -1. Changing r to $\sqrt{2}$ returns zero. The wrong formula has the right phase, and a single evaluation catches it.

The cone uses the harmonic equation. A planar Helmholtz equation with nonzero wavenumber has a different radial potential. Comparing them illustrates what a gauge and a logarithmic coordinate can do in this example. A comparison that isolates the effect of coupling must also hold the differential operator and its spectral parameter fixed.

05 FROM PROOF TO PROGRAM

What an ODE solver can hand back

Give the calculator A, B, C in $Ay'' + By' + Cy = 0$. It first retains the places where an input is undefined or $A = 0$. Cancelling $x - 1$ from all three coefficients may simplify its arithmetic, but cannot turn $x = 1$ into an ordinary point of the input equation. Next it forms $P = B/A$, $Q = C/A$ on the remaining intervals.

For constant P, Q, the characteristic polynomial supplies two modes, including the repeated-root and oscillatory cases. For a shifted Euler equation with $v = x-h$, the identities $vB = pA$, $v^2C = qA$ establish the transformation $s = \log|v|$ and the polynomial $r^2 + (p-1)r + q$. For the other recognised families the program returns the exact slope law, the chart condition and a reason the implemented solution rules stop. A family label alone selects no boundary condition.

Inputs A, B, C	Response, simplified for display
$1, -2, 1$	$y = C_1e^x + C_2xe^x$.
$(1+x)^2, 1+x, -2$	On $1 + x > 0$, modes $(1+x)^{\sqrt{2}}, (1+x)^{-\sqrt{2}}$.
$x^2, x, x^2 - 4$	Bessel order two, $z' = -z^2 - z/x - 1 + 4/x^2$.

Try the calculation

In Better Calculator open *Differential equations*. The three operations cover homogeneous second-order linear equations, autonomous first-order Riccati equations and the cone modes on page 5. The portable Python component uses exact rational coefficients and the standard library. To run the Bessel case, start Python in the calculator repository's `python/` directory.

```
from riccati import solve_linear
result = solve_linear("x^2", "x", "x^2-4")
print(result["slope_rhs"])
```

Run the CAS examples · Read the Lean proof · Build the paper

What was checked

Lean 4.26.0 with mathlib v4.26.0 checks 31 public theorems in the companion. Start with `riccati_iff` for the equivalence and `reconstruct_on` for recovery. The coordinate calculation is `schwarzian_pullback`. The cone calculation is `coupled_mode_solves_iff`. Section numbers in the Lean comments refer to the March draft. Named-function asymptotics come from the references below. The executable calculator is tested separately by substitution, Wronskians and browser runs. The projective equation is common to every regular scalar homogeneous linear equation of order two. Which trajectory a physical experiment follows is still a question for its initial or boundary data.

References

[1] NIST Digital Library of Mathematical Functions, §1.13, Differential Equations.

[2] NIST DLMF, §10.5, Wronskians, §10.7, Limiting Forms, §10.17, Large Argument and §10.21, Zeros.

[3] NIST DLMF, §9.7, Airy Asymptotics and §9.9, Airy Zeros.

[4] R. E. Langer, *On the Connection Formulas and the Solutions of the Wave Equation*, Physical Review 51 (1937), 669–676. doi:10.1103/PhysRev.51.669.